

Rehabilitate Game+: Pose estimation-based serious games for forearm and hand movement rehabilitation

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ABSTRACT

Motor sequelae in the hand and forearm can result from various diseases or injuries. As a result, the patient has limitations or inability to perform movements. Serious games for motor rehabilitation can motivate patients in the exercise sessions needed to recover the compromised limb. This study aimed to extend a game platform, developing and evaluating serious games focused on the rehabilitation of hand and forearm movements, then creating the Rehabilitate Game+. To develop it, a co-design methodology was used with 21 health professionals (16 physiotherapists and 5 psychologists). The result was a list of requirements, which were implemented in two games based on human hand pose estimation. After that, the Rehabilitate Game+ was first evaluated with 21 physiotherapists in a usability and user experience study applying the System Usability Scale (SUS) and User Experience Questionnaire (UEQ). Also, movement recognition techniques employed in the games had performance evaluated with 36 healthy participants. The average score of SUS was 92.5 on a 100-point scale and UEQ results varied between 1.46 and 1.98 for six factors within its benchmark, which varies on a scale from -3 to +3, so representing good/excellent results. Movement recognition techniques demonstrated high performance (F1-score of 0.98 and 0.99 for forearm pronation and supination, respectively, and 1.00 for finger flexion and extension). We conclude that the proposed games can be experimented in clinical trials with patients.

1. Introduction

The upper limbs of the human body play an important role in everyday tasks, both those involving fine motor skills, which require precision and involve smaller muscle groups, and gross motor skills, which require impulse and strength and cover a wider range of muscles. The integrity of upper limb motor function is essential to ensure human functionality, when performing daily tasks, such as lifting objects (e.g., lifting cutlery to eat), holding clothes to get dressed, and practicing physical activities [1]. The involvement of the upper limbs in various tasks is favored by their mobility, whose joints have a large range of motion. The execution of these tasks with adequate muscular strength, dexterity, and bimanual coordination depends on a great integration of sensory and motor stimuli [2].

The upper limbs can be affected by injuries caused by diseases and direct or indirect trauma. In patients who are victims of stroke, for example, 80% of survivors have motor impairment of the upper limbs, of which only 20% to 30% achieve functional recovery [3]. Among

traumatic injuries, fractures are a frequent condition, for example, caused by traffic accidents and sports injuries, which affect people of all ages [4]. In addition, the elbow and wrist are joints frequently affected by injuries associated with repetitive strain, such as Pronator teres syndrome and Carpal tunnel syndrome. Other problems associated with upper limbs include: de Quervain's tenosynovitis, Brachial plexus injury, and Lateral epicondylitis. All of them can have a direct impact on the function of the upper limbs: on the forearms (pronation and supination movements), and hands (flexion and extension movements of the fingers). These movements are very important for several daily activities, in addition to having relevance in several professions that require specific manual movements.

Classical motor rehabilitation is mainly conducted by physiotherapists, either through manual techniques or with the support of training equipment. Treatments are usually conducted in rehabilitation centers, under the supervision of the medical staff, where labor and material resource costs may be high [5]. In addition, patients report that the

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classical form of functional recovery can be boring due to repetitive exercises, triggering demotivation and, hence, disinterest, which compromise the rehabilitation process. Other interventions have been used in the recovery of movements of upper limbs, mainly due to stroke, such as constraint-induced movement therapy, mirror therapy and robotic-assisted therapy [6].

Rehabilitation methods for upper limbs based on computer systems, such as serious games or exergames (terms hereinafter used interchangeably) [7,8], are presented as an option to overcome the disadvantages related to lack of motivation in the classic treatment [9]. Serious games provide an interactive experience that generates high levels of patient engagement during rehabilitation [10]. They have been adopted in physical and cognitive recovery through interactive exercises provided in the immersion of the ludic environment, such as body movements to accomplish tasks assigned by game interface elements [9]. Thus, serious games can motivate patients to perform a prescribed sequence of rehabilitation exercises in an attractive and engaging way, in addition to allowing to alleviate the feeling of tiredness during the exercises [11].

The research gap addressed by this study involves the development of serious games to support physiotherapists and patients in the rehabilitation process of forearm rotation movements (pronation and supination) and finger movements (flexion and extension). This study aimed to extend the Rehabilitate Game platform [12] by developing and evaluating new serious games focused on the rehabilitation of hand and forearm movements, thus creating the Rehabilitate Game+. For this purpose, a co-design methodology was used with 21 physiotherapists, which resulted in a list of requirements. They then were implemented in two exergames for the rehabilitation of hand and forearm movements. Developed games track upper limb movements to allow players to interact with game mechanics. For this purpose, a technique based on Human Pose Estimation (HPE) [13,14], which uses computer vision and machine learning, and a single camera are employed. The Rehabilitate Game+ platform was evaluated by 21 physiotherapists in experiments focused on usability and user experience applying the System Usability Scale (SUS) and User Experience Questionnaire (UEQ). Also, the movement recognition techniques implemented in the games were evaluated by 36 healthy participants.

The remaining of this paper is organized as follows. Section 2 provides a brief overview of the hand and forearm with their movements, and human hand pose estimation. Section 3 describes materials and methods of the study. Section 4 presents the results, while Section 5 discusses them, compares the study with prior work, and acknowledges study limitations. At the end, Section 6 derives conclusions.

2. Background

2.1. Hand and forearm movements

The hand is considered one of humanity's greatest tools. With the evolution over the years, the development and constant refinement of this tool were possible. The hand is composed of a specialized structure that works synchronously, providing precise motor biomechanics and fine tactile senses [15]. The anatomy of the hand is divided into phalanges, bones, metacarpals and carpal bones. Specifically, the hand is made up of 27 bones, including eight carpal bones, five metacarpals, and 14 phalanges [15,16]. Its function is quite complex, with interactions between the strength provided by intrinsic and extrinsic musculature, in addition to the stability provided by ligaments and bone structure, ensuring insertion and fixation for muscles and ligaments [16].

Flexion and extension movements are common in almost all joints in the human body that allow free movement, such as the toes, knees, hips, trunk, shoulders, elbows, wrists, and fingers. In the hand, flexion occurs when the fingers curl towards the palm. This is possible thanks to the flexor muscles of the hand, located in the anterior part of the

forearm, which are connected to the bones of the fingers by tendons that allow flexion. Extension is the movement opposite to flexion, in which the fingers extend and move away from the palm. This is possible thanks to the extensor muscles of the hand, located at the back of the forearm [17]. The interphalangeal joints allow for flexion and extension of the fingers. They are considered ginglymus joints, also known as hinge joints, which allow movements in a single plane, such as a door hinge. The distal interphalangeal joints have a range of motion that ranges from approximately 12° to 31°, while the proximal interphalangeal joints have a range of motion that ranges from 19° to 70° [18].

The forearm is also an integral part of the upper limb, which extends from the elbow to the wrist. It is formed by the radius and ulna bones connected by a strong, fibrous interosseous membrane. This membrane enables the attachment/stability of the forearm bones while pronation and supination occur, as well as providing proximal attachment for some deep muscles of the forearm [19]. In the wrist, there is an important joint for rotational movements called distal radioulnar joint, so named because it is formed by the connection between the radius and ulna bones. The movement that occurs at the elbow and radioulnar joints assists the shoulder in applying force and controlling the placement of the hand in space [19].

Pronation and supination are rotational movements that primarily occur in the forearm. Supination is the rotation that enables to position the palm facing upwards and pronation facing downwards, without moving the shoulder. Joint interactions (elbow and forearm) significantly increase a range of effective hand positioning and movement, allowing greater effectiveness in different activities [20]. The elbow and forearm joints are responsible for a wide range of hand and arm movements, allowing individuals to carry out everyday tasks and complex physical activities.

2.2. Human hand pose estimation

HPE is a computer vision and machine learning technique focused on determining the locations of key body joints or landmarks in a human body, or parts of it, from an image or a video [13]. Different HPE models have been proposed in the literature, which are mainly based on deep learning [14], such as OpenPose, PoseNet, MoveNet, and MediaPipe [21]. The hand detection task via HPE is challenging [22, 23], considering the hand size variation and the need to detect occluded or self-occluded hands. MediaPipe [24] is a framework that provides a library, specifically the *MediaPipe Hands* [25], able to estimate the landmarks of the hand. By using a single camera, *MediaPipe Hands* captures hand gestures from a video stream and recognizes 21 landmarks (i.e., x and y coordinates and z , which is a relative depth) of a hand image in real time, as illustrated in Fig. 1.

In summary, the *MediaPipe Hands* pipeline has two machine learning models [25]: the first one (the palm detector) is applied to the complete image to detect the hand, delimiting it with a bounding box; and the second model (the hand landmark model) works only in the previously bounding box to identify the landmarks. The user's hand gestures, detected from the hand landmarks, are used in the games developed for the Rehabilitate Game+ platform. Of the 21 landmarks identified by the *MediaPipe Hands*, the games use specific coordinates of nine of them (identified in blue in Fig. 1) applying functions to recognize hand movements, as explained in Section 3.3.

3. Materials and methods

Fig. 2 presents a timeline that summarizes the study methodology. Initially, we added two new games to the Rehabilitate Game platform [12], which were focused on exercises for the forearm pronation and supination, and finger flexion and extension. Then, to improve the initial proposal of the developed games, the co-design method was adopted, which involved the collaboration of health professionals.

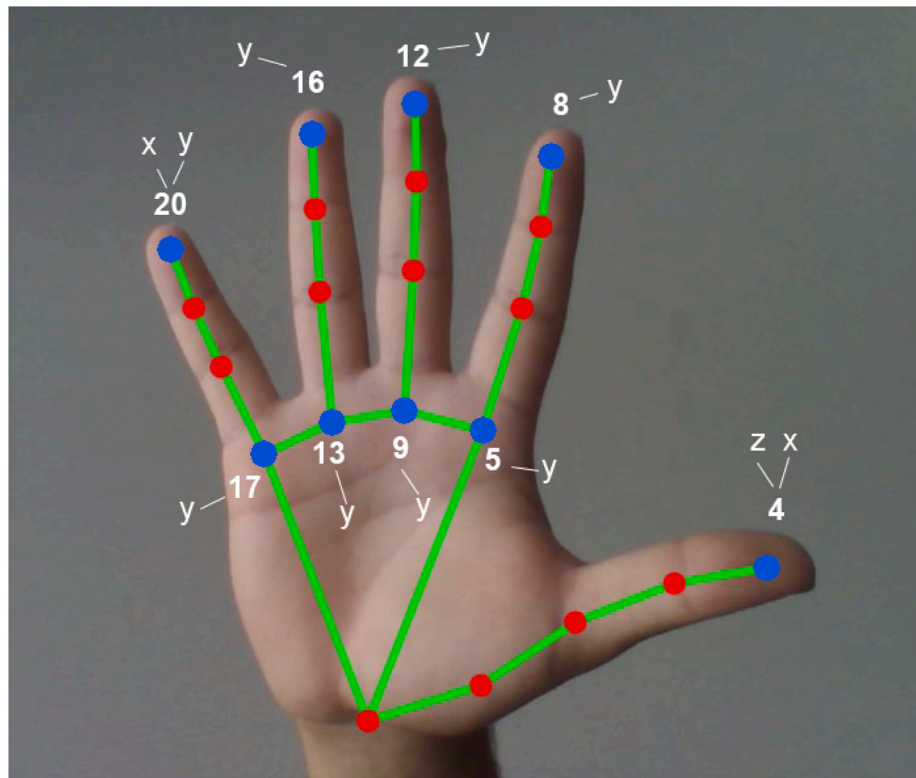


Fig. 1. Tracking 21 landmarks of the hand with the *MediaPipe Hands* solution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

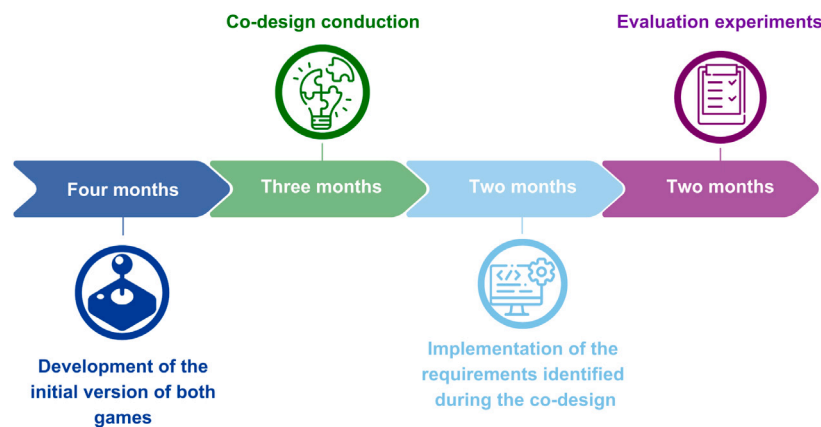


Fig. 2. Research methodology timeline.

They contributed suggestions and ideas, which then generated a list of game requirements. Considering this list, an update of the game implementation was carried out, aiming to improve the platform and the proposed serious games. Rehabilitate Game+ was then created, which was tested in two experiments: the first to evaluate the usability and user experience, involving only physiotherapists, and the second to evaluate the performance in recognizing forearm and hand movements by the techniques employed in the games.

This study was conducted in compliance with the ethical principles inherent in research involving human beings. The study was submitted to the Research Ethics Committee of the Parnaíba Delta Federal University, which was approved under number 5.632.311. Participants were informed about the purpose of the research, its objectives and procedures. After clarification, they were instructed to sign an informed consent form.

3.1. Adding exergames

We added two exergames with exercises based on the gestures of patients/players. The games use a camera to capture user movements, without the need for any other additional input device, such as controls or sensors (see description in Section 3.3). The serious games were developed using the web languages HTML, CSS, JavaScript and PHP, and the Laravel framework. The *MediaPipe Hands* model was incorporated into games to capture the patient's movements and thus allowing control of game elements.

3.1.1. Game 1

In the first game, the scenario is inspired by the adventure of a bee (Video 1 - Supplementary Material), and has a yellow background with beehives and graphic elements, such as flowers, spiders, and a bee. The flower and spider elements descend vertically across the game

screen, and the bee remains at the bottom of the scene. Flowers and spiders appear randomly on the scene. The movement of the bee is controlled by the pronation and supination movements of the patient's forearm, which have three states: neutral (position in which the little finger is positioned laterally to the camera), pronation and supination. Movement control changes depending on the side of the patient's limb, whether left or right.

The patient's objective, when controlling the bee to the left or right, is to catch (i.e., collide) the greatest number of flowers before time runs out. Each flower picked by the patient is counted in their score. Furthermore, the patient must control the bee to avoid spiders in the scenario, using pronation and supination movements. In the web platform, professionals can define the necessary settings for the patient, such as game duration, limb side, game mechanics (i.e., only flowers, only spiders, and both), and speed (i.e., slow, medium, and fast) of the objects (i.e., flower, spider, and bee).

3.1.2. Game 2

The second game has a space-inspired scenario (Video 2 - Supplementary Material), in which the patient's objective is to control a ship, dodge or destroy meteors that descend vertically towards it. The size and position of the meteors are random, so they appear with different sizes and positions across the scenario. The ship only has the possibility of moving horizontally (i.e., left or right). The ship's movement is controlled by the patient's pronation and supination movements, similar to Game 1. In Game 2, players can fire shots from the ship to hit meteors. For this purpose, the shots are performed by flexing and extending the fingers of the hand. Importantly, the shot is only performed by flexing the fingers, so requiring the player to execute the extension movement. Health professionals are able to define the necessary settings for the patient in the web platform, such as the duration of the game, member side, and speed (i.e., slow, medium, and fast) of the objects (i.e., meteors and ship).

3.2. Co-design conduction

The co-design method involves working together to design a new healthcare system, product, and service or update an existing one, leveraging each participant's knowledge, experience, and contributions to achieve better results [26,27]. This co-participation approach has been adopted in serious game projects, since the involvement of end-users and/or those responsible for patient care contributes to the games' performance in their serious purpose [28–31].

3.2.1. Objective

We carried out the co-design with the purpose of improving the two serious games that help in the rehabilitation of upper limbs. This process included the participation of physiotherapists and psychologists. The objective of the collaboration was to understand limitations and needs of patients, in addition to obtaining ideas and suggestions on the functionalities, gamification mechanisms and engagement for the proposed games. Based on the data collected, a list of requirements was generated. The application of the co-design method aimed to make the rehabilitation process more motivating and fun for patients, and to also ensure that the exercises generated from playing the game were effective in rehabilitation.

3.2.2. Participant recruitment

We recruited professionals in the areas of physiotherapy and psychology. The recruitment of participants was initially carried out by invitation to professionals/researchers from the Technological Neuro Innovation & Brain Mapping Laboratory (NitLab) of Parnaíba Delta Federal University (UFDPAr). By first contacting these participants, we carried out snowballing, i.e., invited professionals were initially asked to indicate other ones who could contribute to the study. Thus, recruitment was expanded to other professionals who were not part of the laboratory.

3.2.3. Procedures

Meetings were held with participants at NitLab. At the beginning of the meetings, the web platform and the two added games were explained. The games were then demonstrated and participants had the opportunity to play them. During the demonstration and testing of the games, participants were invited to make suggestions and provide guidance on the platform and games. The meetings lasted an average of 2 h, conducted engagingly, allowing participants to ask questions and share their experiences. The participation of physiotherapists and psychologists allowed a multidisciplinary approach in evaluating the platform and games, enriching the discussion about their usability and effectiveness for patient rehabilitation.

3.2.4. Data collection and analysis

To ensure adequate recording of all data collected, a researcher was assigned to take accurate and detailed notes. With the help of this researcher, it was possible to document all the requirements identified by the participants, ensuring that nothing was missed. At the end, participants answered a questionnaire, one for each group of professionals.

We refined the collected requirements, i.e., similar requirements were joined into a single requirement, reducing duplication of ideas. Then, the requirements were separated into two groups:

- Game Elements (GE): it refers to the visual elements/components, which make up the game graphical interface, such as animation, visual effects, colors, characters, obstacles, and scenario;
- Patient Movements (PM): it deals with the sensitivity and incentives for the movements that patients will execute during the game.

3.3. Improving the games

We then implemented the requirements, making updates to the web platform to better adapt to the new games and improvements in the treatment part, adding more session settings, allowing the professional to configure the games according to the patient's needs. Furthermore, we made the games more flexible for patients, offering a wide variety of formats and preventing the game from becoming monotonous. Based on improvements, we started calling our proposal Rehabilitate Game+.

At this stage, we also adjust and finalize the movement recognition techniques employed in the games. Such techniques are functions based on the positions of the key points provided by the *MediaPipe Hands*, calculated for each frame of the game's video stream. To detect pronation and supination movements of the forearm, Eq. (1) is applied.

$$f(a, b, c) = \begin{cases} \text{pronation movement,} & \text{if } a > b \wedge c = -1 \\ \text{supination movement,} & \text{if } a < b \wedge c = -1 \\ \text{neutral point,} & \text{otherwise} \end{cases} \quad (1)$$

where the parameters are the following key points of the *MediaPipe Hands* (Fig. 1): a is the point 4, x coordinate; b is the point 20, x coordinate; and c is the point 4, z coordinate. When the z coordinate value is -1 , it means that the user's palm is positioned so that it is visible to the camera. The neutral point is when the little finger is oriented horizontally directed at the camera, and the thumb points to the face. This neutral point is important to ensure that the game elements remain stationary during the game.

To detect flexion and extension movements of the hand, Eq. (2) is applied.

$$f(a, b) = \begin{cases} \text{flexion movement,} & \text{if } a > b \\ \text{extension movement,} & \text{otherwise} \end{cases} \quad (2)$$

where the parameters are the following key points (Fig. 1): a may be the points 8, 12, 16 and 20 (y coordinates), depending on the finger defined by the professional on the web platform to configure the game; b may be, respectively, the points 5, 9, 13 and 17 (y coordinates), also depending on the finger defined in the configuration.

3.4. Usability and user experience evaluation

3.4.1. Objective and participant recruitment

We evaluated the usability and user experience in relation to platform changes and game improvements implemented. Selection criteria were: be a physiotherapist; or be in the last period (i.e., semester) of the physiotherapy course and, at the same time, be in the internship period. Recruitment was carried out at the Physiotherapy School Clinic of UFDPAr. Additionally, we perform snowballing.

3.4.2. Procedures

Initially, we presented the web platform and games to the participants. Next, we presented the experiment protocol, which described in detail all the steps and procedures carried out during the experiment. This ensured that participants could understand the experiment, and also had the opportunity to withdraw from participating. Participants went through a learning block to understand the web platform and games, and then completed the protocol, which included steps to: access the web platform, register a new patient, then create a new treatment involving forearm and finger rehabilitation, configure sessions and, finally, play the games. The execution of the protocol lasted approximately 30 min.

3.4.3. Data collection

After carrying out the tests, participants were asked to answer a questionnaire on demographic data, training and experience in rehabilitation (including physiotherapy interns). Next, participants responded to two instruments: (1) SUS [32] composed of 10 statements on a Likert scale from 1 to 5, in which 1 means “completely disagree” and 5 means “completely agree”; and UEQ [33–35] composed of 26 items consisting of pairs of opposites on a 7-point Likert scale, from −3 (fully agree with negative term) to +3 (fully agree with positive term). The terms analyze six factors: attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty.

3.4.4. Data analysis

Participant characterization data were noted to perform descriptive statistical analysis. In the SUS questionnaire, data were tabulated to calculate the score per participant and descriptive statistics, as well as the overall SUS score [32]. The SUS score calculation works as follows: for odd-numbered questions (1, 3, 5, 7, and 9), 1 is subtracted from the participant’s response; if the questions are even (2, 4, 6, 8, and 10), 5 is subtracted from the response. After adjusting the scores for each question, they are summed, and the result is multiplied by 2.5 to obtain the final score, which can range from 0 to 100.

UEQ results were compared with the UEQ benchmark, also presenting values ranging from −3 to +3 [36]. The UEQ benchmark has a reference dataset of 468 studies related to different products (with a total of 21,175 participants) [35,37]. To verify the internal consistency of the study for each UEQ scale, the values obtained for the Cronbach’s Alpha coefficient were evaluated [38]. We analyzed the results with support from the UEQ data analysis tool available in [37].

3.5. Movement recognition evaluation

3.5.1. Objective and participant recruitment

The objective of this experiment was to evaluate the performance of the movement recognition techniques, i.e., the game’s success rate to identify user movements. For this purpose, an adapted version of the game was developed, without counting points or obstacles in the scenario, just with the ship used in Game 2 (Video 3 - Supplementary Material).

Participant recruitment was open to anyone over 18 years and healthy. Furthermore, the snowballing technique was used to maximize the number of participants.

Table 1

Confusion matrix.

		Executed movements		Total
		Positive	Negative	
Recognized movements	Positive	True Positive	False Positive	TP + FP
	Negative	False Negative	True Negative	FN + TN
	Total	TP + FN	FP + TN	

3.5.2. Procedures

We initially demonstrated how movements should be executed, so explaining what each movement controlled in the game. Participants were able to test whether their movements were being captured properly, and try out the game. This learning block lasted until each participant felt confident and comfortable performing the experiment.

When starting the experiment, the game asked the participant to perform the pronation movement 10 times, always returning to the neutral point. Then, the supination movement was performed, also repeated 10 times and, finally, the finger flexion and extension movements, again with 10 repetitions. Between the three movements, there was a pause for participants to read instructions that appeared on the screen. We explained to the participant that they should perform the movements continuously, even if not all movements were detected. The entire procedure for each participant took approximately 10 min.

3.5.3. Data collection

During the procedure, participants performed the movements according to the instructions given on the screen and the researcher counted the number of movements performed. To ensure that no data was missed, we recorded game screenshot videos during the experiment. Videos were created only to collect movements. The version used for the experiment allowed real-time reproduction of the video on the top left side of the screen, showing the movements captured by the camera, which enabled to count movements. After counting, the participant’s movements were compared with the ship’s movements to check whether there was game action. These data were recorded in a table for analyzing performance metrics.

3.5.4. Performance metrics

Data from each participant were recorded in a confusion matrix, as shown in Table 1, with values described as follows.

- True Positive (TP): correct movement recognition by the game for an executed movement;
- True Negative (TN): correct movement recognition by the game for an unexecuted movement. True negative values were not collected because it was not possible to count the number of times the participant did not perform a movement and the game also did not detect it, as there was no occurrence of an event (i.e., new sample) for validation;
- False Positive (FP): recognition of a movement when there was none, so the game misidentified it;
- False Negative (FN): the game did not recognize an executed movement.

We then analyzed the following well-known performance metrics [39]: precision (Eq. (3)), recall (Eq. (4)), and F1-score (Eq. (5)).

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$F1 - score = \frac{2 * (Precision * Recall)}{Precision + Recall} \quad (5)$$

Table 2
Requirements collected from co-design.

ID	Requirement	Description
GE1	Add backgrounds for kids.	Colorful or themed backgrounds, which the professional can configure to attract the attention of child patients.
GE2	Remove spiders from Game 1.	Participants reported discomfort or phobia due to the spider.
GE3	Add ship options to Game 2.	Add a game mechanic that allows ships to evolve, thus preventing the game from becoming monotonous and demotivating the patient.
GE4	Create a game mechanic to allow the ship and bee to reappear in a different position when hit/destroyed.	When the ship or bee is hit/destroyed, it should appear in another random point on the scenario. This sends the message to the player that it has suffered a “damage”.
GE5	Add new game elements to improve gamification.	Add sprites, to represent player evolutions, and animations, such as collisions.
GE6	Add an introductory presentation before starting the game.	Show the ship entering the scenario and positioning itself at the starting location.
GE7	Change game status bar.	Place the game bar at the top to increase the player's focus and not demand their peripheral vision.
GE8	Enable speed control for game elements.	Add controls to allow professionals to set speeds for game elements depending on the patient/player, as he/she may not be able to perform fast movements.
GE9	Diversify game mechanics for game elements.	Create different game mechanics for game elements to vary difficulty levels and rewards.
PM1	Enable the definition of sensitivity levels for movements.	Enable the definition of levels to allow the patient not to force movements that they cannot yet perform.
PM2	Enable the definition of different fingers to recognize the flexion and extension movements of the fingers.	Professionals will be able to choose which finger is considered to recognize movements, thus making the game more personalized to meet the patient's needs.

4. Results

4.1. Co-design

Participants in the co-design study were 16 physiotherapists and 5 psychologists. Physiotherapists were aged between 25 and 59 years old, 12 women, graduated in physiotherapy with experience in the field from 2 to 25 years. Of the 16 physiotherapists, 14 responded that they had already treated patients with impaired pronation and supination movements, and 15 responded that they had already treated patients with impaired flexion and extension movements of the fingers. Only 2 physiotherapists reported that they had already used computer-based technologies to assist in the motor rehabilitation of their patients, and all of them stated that they would use the serious games developed in this study, when properly implemented and evaluated. Psychologists were aged between 31 and 43 years old, 3 women, graduated in psychology with experience ranging from 6 to 18 years.

Table 2 presents the requirements identified in the co-design process, containing: ID, which shows the requirement group as explained in Section 3.2.4, the requirement and a brief description.

4.2. Games after co-design

After the co-design phase (i.e., collection and refinement of requirements), updates were carried out on both the web platform and two games to meet the identified requirements. The platform has been updated to allow the configuration of games to meet requirements

related to the patient's position and movements, such as definitions of sensitivity levels for performing movements (i.e., range of motion) and frequency of appearance of game elements. Additionally, in Game 2, a setting was added to select which finger should be considered during the game to detect a movement, which is useful for targeting rehabilitation to specific areas of the hand. These game definitions allow the creation of several variations of the game, which are useful to meet the specific needs of the patient, enabling progress in the rehabilitation of compromised movements.

Game scenarios (Videos 4 and 5 - Supplementary Material) have also been updated to provide real-time visual feedback to the player. A video on the top left allows players to see their movements in real time, helping them adjust their posture and improve the movement execution. The status bar at the top center of the scene provides important information, such as session time and number of achievements, allowing players to monitor their progress. In Game 2, an evolution system was added to the ships, representing progress in the game, activated through the destruction of meteors.

4.3. Usability and user experience

The usability and user experience study had 21 participants, including physiotherapists ($n = 10$) and students from the UFDPar physiotherapy course ($n = 11$), who had already completed or were in the internship period. They were aged between 23 and 48 (28.95 ± 5.85). Physiotherapists had experience in clinical practice between 4 and 25 years. Among all, 76.2% stated that they had already treated patients' upper limbs.

The SUS score ranges from 0 to 100, with 68 being an acceptable average score in digital health applications [40]. A score below this threshold indicates that the product may have usability issues. Rehabilitate Game+ received an average score of $92.5 (\pm 9.65)$. Even the lowest scores given by participants (Fig. 3) are still above average.

Fig. 4 presents the distribution of UEQ responses, with predominantly positive and neutral evaluations by participants. A few negative values were identified in relation to the perspicuity and novelty scales.

The UEQ benchmark is a reference for interpreting results obtained in previous user experience studies that used the UEQ questionnaire. The attractiveness and stimulation scales (Fig. 5) had averages of 1.98 and 1.82, respectively, indicating excellent performance. These values are among the 10% best results ever obtained in studies using the UEQ. The attractiveness scale measures the overall perception of the system or product, while the stimulation scale evaluates how exciting and motivating that the user perceives when using the system or product.

On the perspicuity, efficiency and novelty scales, the respective averages of 1.76, 1.56 and 1.52 indicate a Good result, in which the benchmark has only 10% better results, and 75% of the results are worse. The perspicuity scale measures how easy it is to get familiar with the proposed games and to learn how to use them. Efficiency shows if the user can solve their tasks quickly without unnecessary effort. Novelty assesses the degree to which the user perceives the system or product as creative, and different from what already exists. Finally, on the dependability scale, an average of 1.46 was obtained, indicating an above average result. At this point, the UEQ benchmark has 25% better results, while 50% results are worse. Dependability scale assesses if users feel in control of the interaction, and accomplishes what he/she wants safely and predictably.

In the experiment, the following Cronbach's Alpha values were obtained for the UEQ scales: attractiveness (0.97), perspicuity (0.94), efficiency (0.95), dependability (0.87), stimulation (0.99), and novelty (0.96). All values indicate high consistency. According to the UEQ analysis tool, when there are three or more inconsistencies in an assessment, the ideal is to remove the participant from the data analysis. However, in this study, only four participants had one or two inconsistencies on some scales, so they were not removed from the analysis.

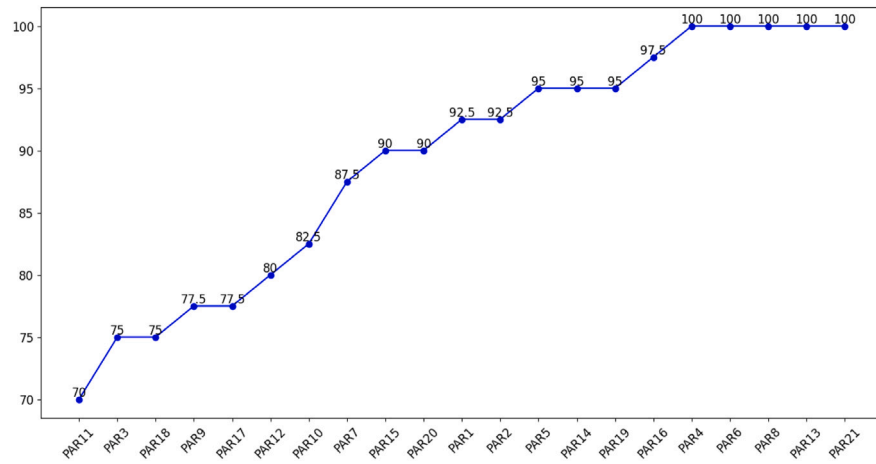


Fig. 3. SUS scores by participant in ascending order.



Fig. 4. Distribution of the 21 participants' responses to the 26 UEQ items. 1 means fully agree with negative item, and 7 means fully agree with positive item.

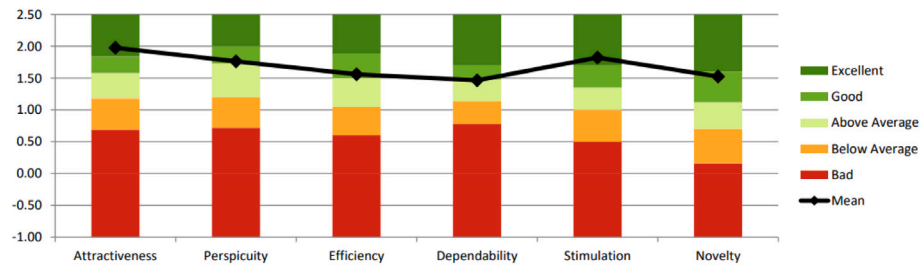


Fig. 5. UEQ benchmark comparing the results of the Rehabilitate Game+ with other software solutions for the six scales.

4.4. Movement recognition

The experimental evaluation of movement recognition included 36 healthy participants of legal age, then generating 36 videos analyzed, and 360 movements of each type performed (i.e., forearm pronation, forearm supination, and finger flexion/extension). The data were tabulated to create confusion matrices, which were analyzed to evaluate the game's ability to detect movements. From the tables, we calculate the precision, recall and F1-score metrics. These calculations were performed for each participant, and the results were then combined to obtain the average metric values for each movement.

Fig. 6(a) shows that the movement detection solution for forearm pronation achieved a precision of 1.0, recall of 0.96, and F1-score of 0.98. We observed a high number of true positives ($n = 347$), indicating that the game correctly identified the occurrence of the movement, and a low number of false negatives ($n = 13$), indicating that there were few cases in which the game did not detect movements, even when they occurred. Fig. 6(b) depicts the performance of the movement detection solution for forearm supination: precision of 1.0, recall of 0.98, and F1-score of 0.99. Also, we observed a high number of true positives ($n = 352$), and a low number of false negatives ($n = 8$). Finally, as displayed in Fig. 6(c), the results of all metrics were 1.0

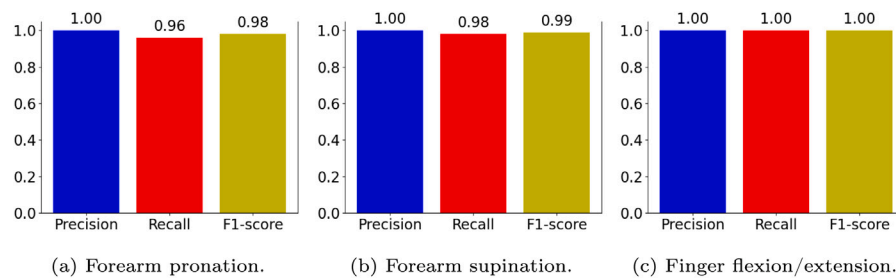


Fig. 6. Performance in detecting movements.

for finger flexion/extension, which means that the movement detection solution recognized the 360 movements of the 36 participants correctly (i.e., true positives). As seen in Fig. 6, there were no false positives for all three movements.

5. Discussion

This work resulted in two serious games integrated into a web platform, allowing the professional to configure them according to the patient's needs. Both games capture patients' movements using a camera and techniques based on a human hand pose estimation model. The first game was created to assist in the rehabilitation of forearm pronation and supination movements, and the second game, in addition to rehabilitating forearm movements, also works on flexion and extension movements of the fingers. The games are tools focused on contributing to the efficiency of the functional recovery of specific compromised movements, in addition to reducing the patient treatment dropout rates.

5.1. Principal findings

The co-design procedure was of great importance for the development of Rehabilitate Game+, as it allowed stakeholders (physiotherapy and psychology professionals) to participate in defining game requirements, thus contributing to empowering the tool to meet the needs and expectations of patients. After gathering the requirements, there were several meetings of the research group to discuss them. They went through a process of analysis and refinement, where similar requirements that had the same objective were merged, resulting in a more concise list. As a result of the discussion process, it was possible to reach an improved version of Rehabilitate Game+. The continuous evolution of the gaming platform can bring benefits to the rehabilitation of patients.

Based on the SUS score, we can say that the usability of the web platform and games performed well, which indicates that the system is easy to use and efficient in execution. This demonstrates that Rehabilitate Game+ was well received by participants in terms of usability. UEQ results indicate that users had a positive experience with the platform and games, with all results above average in the UEQ benchmark. However, there are still opportunities for improvements in some aspects, mainly in the dependability aspect. All of these results will be considered for future versions of web platform and games to improve user experience.

The movement recognition evaluation indicated that the developed techniques presented a satisfactory result for pronation and supination of the forearm, and flexion and extension of the fingers. The results for the precision, recall and F1-score metrics for each movement were high. During the experiments, some reasons were observed why the game was unable to recognize some movements, mainly for pronation and supination movements. We found that the participant's distance from the camera was a factor that made recognition difficult. In some repetitions, participants were too close or too far from the camera. From this evaluation, we were able to identify the reason for incorrect recognitions, so we can avoid them in the future.

5.2. Comparison with prior work

Several serious games and platforms for upper limb rehabilitation have been proposed in the literature [41–45], including those that utilize hardware like Kinect, Nintendo Wii controllers, or robotic devices to detect user movements [46–48]. These solutions often deliver precise motion tracking and adaptive feedback but tend to involve high costs, require specific equipment, and may be limited to clinic-based environments. Additionally, many of them focus broadly on upper limb rehabilitation without targeting specific forearm or hand movements, such as pronation, supination, or finger flexion/extension.

In contrast, Rehabilitate Game+ stands out for its combination of specificity, accessibility, and technological simplicity. By leveraging HPE with *MediaPipe Hands* and a standard webcam, the platform removes the need for specialized devices, making it significantly more affordable and scalable than systems dependent on proprietary hardware. Importantly, the potential of HPE models has already been recognized in the health domain [49] and to meet physiotherapy requirements [50]. Unlike prior vision-based systems [51–58], which generally focus on movement analysis [59] or assessment [60], our approach actively integrates gesture detection into interactive serious games tailored to the motor rehabilitation process.

Moreover, while commercial games may be appealing and user-friendly, they lack therapeutic specificity and adaptability required for rehabilitation [9]. On the other hand, existing therapeutic games are rarely developed using a co-design methodology involving healthcare professionals from the outset [61]. Rehabilitate Game+ benefits from being co-designed with physiotherapists and psychologists, ensuring that the exercises are clinically relevant, customizable, and aligned with therapeutic goals.

Recently, we developed the original version of Rehabilitate Game, a gesture-based platform for upper limb rehabilitation supported by HPE models [12]. Designed as a complementary therapy tool, it enabled playful and manageable motor exercises in both clinical and home settings. The present study introduces Rehabilitate Game+, an enhanced version of the platform that incorporates new serious games specifically developed to support the rehabilitation of forearm pronation/supination and finger flexion/extension, which are two key movement groups not previously addressed. These games were co-designed with healthcare professionals, ensuring alignment with therapeutic practices and patient needs.

Another strength of the Rehabilitate Game+ lies in its scalability and cost-effectiveness. By eliminating the need for robotic devices, depth cameras, or wearable sensors, and relying instead on a standard webcam and browser-based interface, Rehabilitate Game+ reduces costs and simplifies implementation. This makes it viable for widespread use (from clinical rehabilitation centers to patients' homes) especially in underserved or resource-constrained environments. Moreover, the web-based design facilitates remote updates and professional oversight, supporting broad and sustainable integration into diverse healthcare contexts.

Therefore, what distinguishes Rehabilitate Game+ from existing solutions is its combination of (1) HPE-based gesture tracking via a standard

webcam, (2) personalized and configurable gameplay mechanics, and (3) a fully web-based architecture. To the best of our knowledge, it is the first HPE-based serious game platform focused specifically on the rehabilitation of forearm and hand movements, filling a gap left by prior work that often targeted broader motor skills or required expensive, specialized hardware.

5.3. Challenges, limitations, and future work

The proposed system demonstrates significant potential as a versatile tool for motor rehabilitation, with practical implications for both clinical and home-based settings. Its gamification elements, including goal setting, rewards, and adaptive progression, provide an engaging platform to motivate patients during their rehabilitation journey. The ability to adjust game parameters, such as difficulty levels and range of motion requirements, offers a degree of customization that enables therapists to tailor interventions to individual patient needs. These features align well with modern rehabilitation practices [62], emphasizing personalized and patient-centered care.

However, some challenges must be addressed to maximize the system's clinical impact. A key challenge lies in ensuring its adaptability to patients with diverse levels of injury and at different stages of recovery. While the system allows for customization, its effectiveness across various clinical contexts (e.g., acute, subacute, chronic) has yet to be evaluated. Additionally, future experiments may reveal the need for further refinements to ensure the platform effectively accommodates patients across a wide spectrum of disabilities, from mild to severe, maintaining its relevance and usability for all users.

Maintaining long-term patient engagement and motivation represents another critical challenge in serious games for rehabilitation [63]. Even with gamification, sustaining motivation over extended rehabilitation periods can be difficult. The system's current design provides foundational elements to encourage engagement. However, long-term effectiveness in real-world scenarios remains understudied. Future research will aim to evaluate the long-term impact of gamification strategies on enhancing patient engagement and improving clinical outcomes.

Finally, this work has limitations that need to be acknowledged. One of the limitations is the number of participants who contributed to the research in the three stages in which human beings participated (i.e., co-design, usability and user experience evaluation, and movement recognition evaluation), which could be greater. Regarding movement detection, the solution developed is based on rules (i.e., functions), which uses positions between *MediaPipe Hands* key points, and may not generalize all movements well. Therefore, a machine learning-based solution could be more suitable, and should be developed in future studies.

6. Conclusions

This research extended a game platform by adapting it and adding to it two serious gesture-based games for the rehabilitation of hand and forearm movements. The games use a VC solution to capture the patient's gestures through a camera incorporated into a web platform. *Rehabilitate Game+* is played by the patient according to the settings defined by healthcare professionals, which are defined in the personalized treatment for each patient. To keep the patient engaged in the playful environment and, at the same time, in the rehabilitation process, gamification techniques were used, with interactions between the patient and the game elements, providing rewards for patients.

An important contribution of this study was the redefinition and evolution of games through a co-design process, in which the requirements suggested by participants were accepted. After these improvements, *Rehabilitate Game+* went through a period of experiments and showed promising results in evaluating usability and user experience.

These results demonstrate the relevance of the study and its contribution to the development of games aimed at rehabilitation, which meet user needs and preferences. Therefore, we believe that the games developed are capable of being experimented in clinical trials. Plans for future work include carrying out these studies with patients.

CRedit authorship contribution statement

José Everton Fontenele: Writing – review & editing, Validation, Investigation, Conceptualization, Writing – original draft, Software, Data curation, Visualization, Methodology, Formal analysis. **Silmar Teixeira:** Supervision, Project administration, Investigation, Methodology, Conceptualization. **Daniel Lima Sousa:** Writing – original draft, Methodology, Software, Investigation. **Rayele Moreira:** Methodology, Investigation, Data curation. **Maria Gislene Silva:** Investigation, Data curation. **Samaritana Nascimento:** Data curation, Investigation. **Leynilson Pereira:** Methodology, Data curation. **Ariel Soares Teles:** Writing – original draft, Methodology, Supervision, Funding acquisition, Writing – review & editing, Project administration, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.entcom.2025.100992>.

Data availability

No data was used for the research described in the article.

References

- [1] L.A. Ingram, A.A. Butler, L.D. Walsh, M.A. Brodie, S.R. Lord, S.C. Gandevia, The upper limb Physiological Profile Assessment: Description, reliability, normative values and criterion validity, *PLOS ONE* 14 (6) (2019) e0218553, <http://dx.doi.org/10.1371/journal.pone.0218553>.
- [2] L.A. Ingram, A.A. Butler, M.A. Brodie, P. Hoang, S.C. Gandevia, S.R. Lord, Quantifying upper-limb motor impairment in people with multiple sclerosis: A physiological profiling approach, *Ann. Phys. Rehabil. Med.* 65 (5) (2022) 101625, <http://dx.doi.org/10.1016/j.rehab.2021.101625>.
- [3] I. Ahmed, I. Yeldan, R. Mustafaoglu, The adjunct of electric neurostimulation to rehabilitation approaches in upper limb stroke rehabilitation: A systematic review with network meta-analysis of randomized controlled trials, *Neuromodulation: Technol. Neural Interface* (2022) <http://dx.doi.org/10.1016/j.neurom.2022.01.005>.

- [4] M. Michielsen, A. Van Haver, M. Vanhees, R. van Riet, F. Verstrecken, Use of three-dimensional technology for complications of upper limb fracture treatment, *EFORT Open Rev.* 4 (6) (2019) 302–312, <http://dx.doi.org/10.1302/2058-5241.4.180074>.
- [5] W. Ying, W. Aimin, Augmented reality based upper limb rehabilitation system, in: 2017 13th IEEE International Conference on Electronic Measurement & Instruments, ICEMI, IEEE, 2017, pp. 426–430, <http://dx.doi.org/10.1109/ICEMI.2017.8265843>.
- [6] J. Pulman, E. Buckley, Assessing the efficacy of different upper limb hemiparesis interventions on improving health-related quality of life in stroke patients: a systematic review, *Top. Stroke Rehabil.* 20 (2) (2013) 171–188, <http://dx.doi.org/10.1310/tsr2002-171>.
- [7] V. Wattanasoontorn, I. Boada, R. García, M. Sbert, Serious games for health, *Entertain. Comput.* 4 (4) (2013) 231–247, <http://dx.doi.org/10.1016/j.entcom.2013.09.002>.
- [8] G. Chan, A. Arya, R. Orji, Z. Zhao, Motivational strategies and approaches for single and multi-player exergames: a social perspective, *PeerJ Comput. Sci.* 5 (2019) e230, <http://dx.doi.org/10.7717/peerj-cs.230>.
- [9] T. Kamnarsiri, K. Phiro, S. Boripuntakul, S. Sungkarat, An interactive physical-cognitive game-based training system using Kinect for older adults: Development and usability study, *JMIR Serious Games* 9 (4) (2021) e27848, <http://dx.doi.org/10.2196/27848>.
- [10] Y. Chen, K.T. Abel, J.T. Janeczek, Y. Chen, K. Zheng, S.C. Cramer, Home-based technologies for stroke rehabilitation: A systematic review, *Int. J. Med. Inform.* 123 (2019) 11–22, <http://dx.doi.org/10.1016/j.ijmedinf.2018.12.001>.
- [11] T. Alves, R.S. Gonçalves, G. Carbone, Serious games strategies with cable-driven robots for bimanual rehabilitation: A randomized controlled trial with Post-Stroke patients, *Front. Robot. AI* 9 (2022) 739088, <http://dx.doi.org/10.3389/frobt.2022.739088>.
- [12] R. Baluz, A. Teles, J.E. Fontenele, R. Moreira, R. Fialho, P. Azevedo, D. Sousa, F. Santos, V.H. Bastos, S. Teixeira, Motor rehabilitation of upper limbs using a gesture-based serious game: Evaluation of usability and user experience, *Games Heal. J.* (2022) <http://dx.doi.org/10.1089/g4h.2022.0005>.
- [13] S. Dubey, M. Dixit, A comprehensive survey on human pose estimation approaches, *Multimedia Syst.* 29 (2023) 167–195, <http://dx.doi.org/10.1007/s00530-022-00980-0>.
- [14] C. Zheng, W. Wu, C. Chen, T. Yang, S. Zhu, J. Shen, N. Kehtarnavaz, M. Shah, Deep learning-based human pose estimation: A survey, *ACM Comput. Surv.* (2023) <http://dx.doi.org/10.1145/3603618>.
- [15] J. Maw, K.Y. Wong, P. Gillespie, Hand anatomy, *Br. J. Hosp. Med.* 77 (3) (2016) C34–C40, <http://dx.doi.org/10.12968/hmed.2016.77.3.C34>.
- [16] S. Panchal-Kildare, K. Malone, Skeletal anatomy of the hand, *Hand Clin.* 29 (4) (2013) 459–471, <http://dx.doi.org/10.1016/j.hcl.2013.08.001>.
- [17] J. Hamill, K. Knutzen, T. Derrick, *Biomechanical Basis of Human Movement*, LWW, 2021.
- [18] S. Hall, *Basic Biomechanics*, vol. 7, McGraw-Hill, 2014.
- [19] K.L. Moore, A.F. Dalley, A.M.R. Agur, *Clinically Oriented Anatomy*, Wolters Kluwer Health/Lippincott Williams & Wilkins, 2010.
- [20] D.A. Neumann, *Kinesiology of the Musculoskeletal System: Foundations for Rehabilitation*, vol. 3rd, Elsevier, 2016.
- [21] J.-L. Chung, L.-Y. Ong, M.-C. Leow, Comparative analysis of skeleton-based human pose estimation, *Futur. Internet* 14 (12) (2022) <http://dx.doi.org/10.3390/fi14120380>.
- [22] D. Sarma, M. Bhuyan, Methods, databases and recent advancement of vision-based hand gesture recognition for HCI systems: A review, *SN Comput. Sci.* 2 (2021) <http://dx.doi.org/10.1007/s42979-021-00827-x>.
- [23] M. Oudah, A. Al-Naji, J. Chahl, Hand gesture recognition based on computer vision: A review of techniques, *J. Imaging* 6 (8) (2020) <http://dx.doi.org/10.3390/jimaging6080073>.
- [24] C. Lugaesi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M.G. Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, M. Grundmann, MediaPipe: A framework for building perception pipelines, 2019, <http://dx.doi.org/10.48550/arXiv.1906.08172>, arXiv:1906.08172, URL <http://arxiv.org/abs/1906.08172>.
- [25] F. Zhang, V. Bazarevsky, A. Vakunov, A. Tkachenka, G. Sung, C.-L. Chang, M. Grundmann, Mediapipe hands: On-device real-time hand tracking, 2020, <http://dx.doi.org/10.48550/arXiv.2006.10214>, arXiv preprint arXiv:2006.10214.
- [26] M. Bird, M. McGillion, E. Chambers, J. Dix, C. Fajardo, M. Gilmour, K. Levesque, A. Lim, S. Mierdel, C. Ouellette, A. Polanski, S. Reaume, C. Whitmore, N. Carter, A generative co-design framework for healthcare innovation: development and application of an end-user engagement framework, *Res. Involv. Engag.* 7 (2021) <http://dx.doi.org/10.1186/s40900-021-00252-7>.
- [27] P. Slattery, A. Saeri, P. Bragge, Research co-design in health: A rapid overview of reviews, *Heal. Res. Policy Syst.* 18 (2020) <http://dx.doi.org/10.1186/s12961-020-0528-9>.
- [28] M. Pirovano, E. Surer, R. Mainetti, P.L. Lanzi, N. Alberto Borghese, Exergaming and rehabilitation: A methodology for the design of effective and safe therapeutic exergames, *Entertain. Comput.* 14 (2016) 55–65, <http://dx.doi.org/10.1016/j.entcom.2015.10.002>.
- [29] M.R. Dekker, A.D. Williams, The use of user-centered participatory design in serious games for anxiety and depression, *Games Heal. J.* 6 (6) (2017) 327–333, <http://dx.doi.org/10.1089/g4h.2017.0058>.
- [30] M. Ward, É. Ní Shé, A. De Brún, C. Korpos, M. Hamza, E. Burke, A. Duffy, K. Egan, U. Geary, C. Holland, et al., The co-design, implementation and evaluation of a serious board game ‘PlayDecide patient safety’ to educate junior doctors about patient safety and the importance of reporting safety concerns, *BMC Med. Educ.* 19 (1) (2019) 1–14, <http://dx.doi.org/10.1186/s12909-019-1655-2>.
- [31] T. Alves, H. Carvalho, D. Simões Lopes, Winning compensations: Adaptable gaming approach for upper limb rehabilitation sessions based on compensatory movements, *J. Biomed. Inform.* 108 (2020) 103501, <http://dx.doi.org/10.1016/j.jbi.2020.103501>.
- [32] J. Brooke, SUS: A ‘quick’ and ‘dirty’ usability scale, in: P.W. Jordan, B. Thomas, B.A. Weerdmeester, L.L. McClelland (Eds.), *Usability Evaluation in Industry*, Taylor and Francis, 1996, pp. 189–194.
- [33] B. Laugwitz, T. Held, M. Schrepp, Construction and evaluation of a user experience questionnaire, in: A. Holzinger (Ed.), *HCI and Usability for Education and Work*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2008, pp. 63–76, http://dx.doi.org/10.1007/978-3-540-89350-9_6.
- [34] M. Schrepp, A. Hinderks, J. Thomaschewski, Applying the user experience questionnaire (UEQ) in different evaluation scenarios, in: A. Marcus (Ed.), *Design, User Experience, and Usability. Theories, Methods, and Tools for Designing the User Experience*, Springer International Publishing, Cham, 2014, pp. 383–392, http://dx.doi.org/10.1007/978-3-319-07668-3_37.
- [35] A. Hinderks, M. Schrepp, F.J. Domínguez Mayo, M.J. Escalona, J. Thomaschewski, Developing a UX KPI based on the user experience questionnaire, *Comput. Stand. Interfaces* 65 (2019) 38–44, <http://dx.doi.org/10.1016/j.csi.2019.01.007>.
- [36] M. Schrepp, J. Thomaschewski, A. Hinderks, Construction of a benchmark for the user experience questionnaire (UEQ), *Int. J. Interact. Multimed. Artif. Intell.* 4 (4) (2017) 40–44, <http://dx.doi.org/10.9781/ijimai.2017.445>.
- [37] A. Hinderks, M. Schrepp, J. Thomaschewski, User experience questionnaire, 2023, Available on: <https://www.ueq-online.org>.
- [38] J.M. Bland, D.G. Altman, Statistics notes: Cronbach’s alpha, *BMJ* 314 (7080) (1997) 572, <http://dx.doi.org/10.1136/bmj.314.7080.572>.
- [39] M. Sokolova, G. Lapalme, A systematic analysis of performance measures for classification tasks, *Inf. Process. Manage.* 45 (4) (2009) 427–437, <http://dx.doi.org/10.1016/j.ipm.2009.03.002>.
- [40] M. Hyzy, R. Bond, M. Mulvenna, L. Bai, A. Dix, S. Leigh, S. Hunt, System usability scale benchmarking for digital health apps: Meta-analysis, *JMIR Mhealth Uhealth* 10 (8) (2022) e37290, <http://dx.doi.org/10.2196/37290>.
- [41] J.P. Proença, C. Quaresma, P. Vieira, Serious games for upper limb rehabilitation: a systematic review, *Disabil. Rehabil. Assist. Technol.* 13 (1) (2018) 95–100, <http://dx.doi.org/10.1080/17483107.2017.1290702>.
- [42] E. Koutsiana, I. Ladakis, D. Fotopoulos, A. Chytas, V. Kilintzis, I. Chouvarda, et al., Serious gaming technology in upper extremity rehabilitation: scoping review, *JMIR Serious Games* 8 (4) (2020) e19071, <http://dx.doi.org/10.2196/19071>.
- [43] I. Dumas, G. Everard, S. Dehem, T. Lejeune, Serious games for upper limb rehabilitation after stroke: a meta-analysis, *J. Neuroeng. Rehabil.* 18 (1) (2021) 1–16, <http://dx.doi.org/10.1186/s12984-021-00889-1>.
- [44] M.F. Pereira, C. Prahm, J. Kolbenschlag, E. Oliveira, N.F. Rodrigues, Application of AR and VR in hand rehabilitation: A systematic review, *J. Biomed. Inform.* 111 (2020) 103584, <http://dx.doi.org/10.1016/j.jbi.2020.103584>.
- [45] R. Kabir, M.S.H. Sunny, H.U. Ahmed, M.H. Rahman, Hand rehabilitation devices: A comprehensive systematic review, *Micromachines* 13 (7) (2022) <http://dx.doi.org/10.3390/mi13071033>.
- [46] C.A. Aguilar-Lazcano, E.J. Rechy-Ramirez, H. Hu, H.V. Rios-Figueroa, A. Marin-Hernandez, Interaction modalities used on serious games for upper limb rehabilitation: a systematic review, *Games Heal. J.* 8 (5) (2019) 313–325, <http://dx.doi.org/10.1089/g4h.2018.0129>.
- [47] M.A. Padilla-Castañeda, E. Sotgiu, A. Frisoli, M. Bergamasco, P. Orsini, A. Martiradonna, S. Olivieri, G. Mazzinghi, C. Laddaga, A virtual reality system for robotic-assisted orthopedic rehabilitation of forearm and elbow fractures, in: 2013 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2013, pp. 1506–1511, <http://dx.doi.org/10.1109/IROS.2013.6696548>.
- [48] M. Castañeda, E. Sotgiu, M. Barsotti, A. Frisoli, O. Piero, A. Martiradonna, C. Laddaga, M. Bergamasco, An orthopaedic robotic-assisted rehabilitation method of the forearm in virtual reality physiotherapy, *J. Heal. Eng.* 2018 (2018) 1–20, <http://dx.doi.org/10.1155/2018/7438609>.
- [49] J. Stenum, K.M. Cherry-Allen, C.O. Pyles, R.D. Reetzke, M.F. Vignos, R.T. Roem-mich, Applications of pose estimation in human health and performance across the lifespan, *Sensors* 21 (21) (2021) <http://dx.doi.org/10.3390/s21217315>.
- [50] T. Hellstén, J. Karlsson, M. Shamsuzzaman, G. Pulkakis, The potential of computer vision-based marker-less human motion analysis for rehabilitation, *Rehabil. Process. Outcome* 10 (2021) 117957272110223, <http://dx.doi.org/10.1177/11795727211022330>.
- [51] G. Prange, T. Krabben, B. Molier, H. van der Kooij, M. Jannink, A low-tech virtual reality application for training of upper extremity motor function in neurorehabilitation, in: 2008 Virtual Rehabilitation, IEEE, 2008, pp. 8–12, <http://dx.doi.org/10.1109/ICVR.2008.4625113>.

- [52] J.W. Burke, M. McNeill, D. Charles, P. Morrow, J. Crosbie, S. McDonough, Serious games for upper limb rehabilitation following stroke, in: 2009 Conference in Games and Virtual Worlds for Serious Applications, IEEE, 2009, pp. 103–110, <http://dx.doi.org/10.1109/VS-GAMES.2009.17>.
- [53] J.W. Burke, M.D.J. McNeill, D.K. Charles, P.J. Morrow, J.H. Crosbie, S.M. McDonough, Optimising engagement for stroke rehabilitation using serious games, *Vis. Comput.* 25 (12) (2009) 1085–1099, <http://dx.doi.org/10.1007/s00371-009-0387-4>.
- [54] G. Alankus, A. Lazar, M. May, C. Kelleher, Towards customizable games for stroke rehabilitation, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, 2010, pp. 2113–2122, <http://dx.doi.org/10.1145/1753326.1753649>.
- [55] L. Evett, A. Burton, S. Battersby, D. Brown, N. Sherkat, G. Ford, H. Liu, P. Standen, Dual camera motion capture for serious games in stroke rehabilitation, in: 2011 IEEE 1st International Conference on Serious Games and Applications for Health, SeGAH, IEEE, 2011, pp. 1–4, <http://dx.doi.org/10.1109/SeGAH.2011.6165460>.
- [56] A.M. Burton, H. Liu, S. Battersby, D. Brown, N. Sherkat, P. Standen, M. Walker, The use of motion tracking technologies in serious games to enhance rehabilitation in stroke patients, *Int. J. Game-Based Learn. (IJGBL)* 1 (4) (2011) 60–73, <http://dx.doi.org/10.4018/ijgbl.2011100106>.
- [57] J. Lin, C.L. Kelleher, J.R. Engsberg, Developing home-based virtual reality therapy interventions, *GAMES HEALTH: Res. Dev. Clin. Appl.* 2 (1) (2013) 34–38, <http://dx.doi.org/10.1089/g4h.2012.0033>.
- [58] A. Jaume-i Capó, P. Martínez-Bueso, B. Moyà-Alcover, J. Varona, Interactive rehabilitation system for improvement of balance therapies in people with cerebral palsy, *IEEE Trans. Neural Syst. Rehabil. Eng.* 22 (2) (2013) 419–427, <http://dx.doi.org/10.1109/TNSRE.2013.2279155>.
- [59] B. Debnath, M. O'Brien, M. Yamaguchi, A. Behera, A review of computer vision-based approaches for physical rehabilitation and assessment, *Multimedia Syst.* 28 (2021) <http://dx.doi.org/10.1007/s00530-021-00815-4>.
- [60] R. Moreira, R. Fialho, A.S. Teles, V. Bordalo, S.S. Vasconcelos, G.P. de Moraes Gouveia, V.H. Bastos, S. Teixeira, A computer vision-based mobile tool for assessing human posture: A validation study, *Comput. Methods Programs Biomed.* 214 (2022) 106565, <http://dx.doi.org/10.1016/j.cmpb.2021.106565>.
- [61] D.L. Sousa, S. Teixeira, J.E. Fontenele, R.M. Santos, L. Pereira, R. Baluz, V.H. Bastos, A.S. Teles, Health care professional-supported co-design of a mime therapy-based serious game for facial rehabilitation, *JMIR Serious Games* 12 (2024) e52661, <http://dx.doi.org/10.2196/52661>.
- [62] J.G. Owens, M.R. Rauzi, A. Kittelson, J. Graber, M.J. Bade, J. Johnson, D. Nabhan, How new technology is improving physical therapy, *Curr. Rev. Musculoskelet. Med.* 13 (2) (2020) 200–211, <http://dx.doi.org/10.1007/s12178-020-09610-6>.
- [63] D.L. Sousa, S. Teixeira, A.C. de Oliveira, L.D.B. Lima, R. Moreira, F. Silva, A.S. Teles, Computer-based gamified tools for facial rehabilitation: A scoping review, *Games Heal. J.* (2024) <http://dx.doi.org/10.1089/g4h.2023.0220>.